Justin Pham

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Education

Oregon State University *Corvallis, OR*
M.Eng in Computer Science, GPA 4.0/4.0 *Sept 2024 – December 2026*
B.S. in Computer Science, GPA 3.99/4.0 *Sept 2021 – December 2025*

Coursework: Operating Systems, High Performance Computer Architecture, Computer Networks, Digital Logic Design

Skills

Languages & Frameworks: Python, C, C++, CUDA, TypeScript, React, SQL, Shell Scripting
Technologies & Tools: Linux, Docker, GitHub Actions, CI/CD, CMake, Makefile, Git, GDB, CppUTest, SQLite, PostgreSQL

Experience

Skydio *(Future) Sept 2026 - Dec 2026*
Incoming System Software Engineer Intern *San Mateo, CA*

- Incoming Software Engineer Intern on Skydio's Middleware team

AMD *Jun 2026 - Present*
System Software Engineer Intern *Austin, TX*

- Leading development of a VS Code extension in TypeScript, React, and Node.js to visualize unit test coverage
- Built an AI agent to integrate register changes, fix builds, and open PRs, saving 80 hours of developer work per task
- Owned validation efforts for EDK2 kernel integration for various client platforms resolving build and boot issues

Qumulo *Mar 2026 - Jun 2026*
Software Engineer Intern *Seattle, WA*

- Built socket connection telemetry in C and Linux to report latency and usage, aiding customer support blind spots
- Decreased customer workload time by 50% by building a prefetcher in C to fetch metadata on NFS/SMB file lookups
- Architected cluster consensus telemetry with RabbitMQ, Grafana, SQL, and InfluxDB to debug performance regressions

AMD *Jan 2025 - Aug 2025*
System Software Engineer Intern *Austin, TX*

- Ported Windows/MSVC test infrastructure to Linux/GCC, reducing CI/CD runtime from 4 hours to 10 minutes
- Built an MCP server exposing tools for testing, and led discussions and meetings to gain adoption by BIOS teams
- Sped up unit test dispatch time by 3x using a thread pool to parallelize execution and aggregate results
- Aided CI deployment of a GitHub Actions unit test check for BIOS platforms, guiding architectural decisions

Intel *Apr 2024 - Sep 2024*
System Software Engineer Intern *Hillsboro, OR*

- Developed a Python tool predicting UEFI driver boot flows, rendering results as PlantUML boot-flow diagrams
- Engineered a Python parser for EDK2 .ini files to extract driver metadata across 3,500 code files in 1 minute
- Profiled Python performance with line_profiler to identify 17x speedup operation using directory-mapping cache

Projects

JOS Operating System | *C*

- Created OS subsystems including the bootloader, virtual memory with page tables, and user-level environments
- Developed preemptive multitasking via clock interrupts, round-robin scheduling, and multiprocessor synchronization

Embedded Temperature Logger | *C, STM32, SPI, GPIO, DMA*

- Implemented a bare-metal DHT sensor driver on an STM32L476RG using GPIO to output temperature readings
- Engineered an SPI LCD driver, including an ASCII-to-glyph rendering pipeline to display temperature data

HTTP Server | *C, Makefile, Linux*

- Built an HTTP/1.1 server in C using TCP sockets to serve web content with keep-alive connections and timeouts
- Optimized request handling with epoll I/O multiplexing and zero-copy sendfile, achieving increased throughput

Correlation Power Analysis Side Channel Attack | *Python, NumPy, Jupyter Notebook*

- Built a side-channel analysis tool in Python to recover AES-128 secret keys from hardware power traces
- Optimized correlation computation using Zarr compression and NumPy from an 8GB dataset in ~2 minutes